BikeShare Commute vs. Leisure Use Across Four Metro Areas

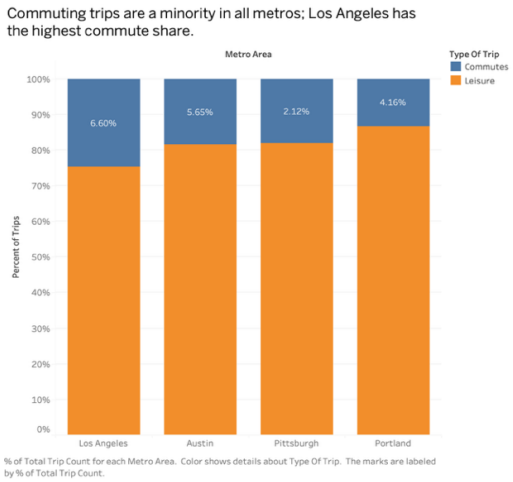
Refined analytic question: *What share of trips in each metro are "commute" vs "leisure" using a consistent rule-based classifier, and which metro is most commute heavy?*

Commute trips are a minority in all four metros; Los Angeles has the highest commute share (24.8%) and Portland the lowest (13.4%) under the stated classifier. BikeShare trip purpose isn't recorded, so I classify "commute" trips using consistent timing, duration, one-way criteria, and compare commute vs. leisure patterns across metros. A trip is labeled "commute" if it meets the following criteria, otherwise it's a "leisure":

- If it is a weekday (Monday through Friday)
- If the hour it starts it either between 6 AM and 9 AM or 4 PM and 6 PM
- If the duration of the trip is between 3 and 30 minutes.
- If it is a one-way trip.
- Excluding dockless from my analysis (From/ToStationID can't equal -5000)

Metrics: Trip counts by metro and type; commute share = CommuteTrips / (CommuteTrips + LeisureTrips).

Insight 1: "Commute" trips are a minority in every metro (highest in LA, lowest in Portland).
Commute shares by metro: Los Angeles 24.8%**,** Austin 18.5%**,** Pittsburgh 18.1%**,** Portland 13.4%**.**



Insight 2: "Commute-heavy" differs from "commute volume"; the table shows the underlying counts. Los Angeles has the highest commute share**,** while Portland has substantial commute counts but is dominated by leisure volume, lowering its share.

Trip Counts by Metro and Trip Type (Commute vs Leisure)

	Type Of Trip	
Metro Area	Leisure	Commutes
Los Angeles	829,157	272,911
Austin	1,030,300	233,528
Pittsburgh	397,750	87,891
Portland	1,113,028	171,884

Sum of Trip Count broken down by Type Of Trip vs. Metro Area.

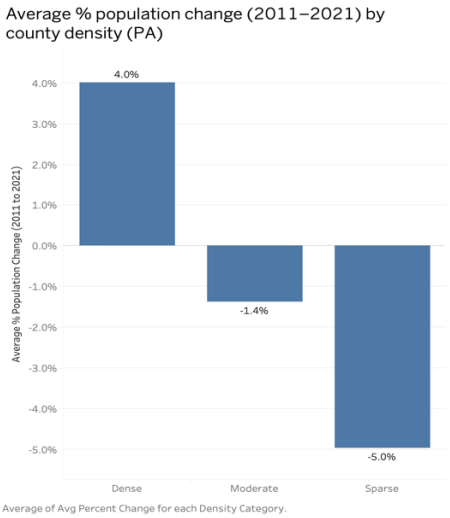
US Demographics: Population Change (2011–2021) by County Density in Pennsylvania

Refined analytic question: *How does average percent population change (2011 to 2021) differ across PA counties classified as dense vs. moderate vs. sparse***?**

Population change was not proportional across density categories, **as dense counties grew on average (+4.0%)**, **while moderate (−1.4%) and sparse counties (−5.0%) declined, indicating systematically different proportional change by density.** The method (density + classification) I used is density_2021 = pop2021 / land_sq_miles, where pop20xx and land_sq_miles are summed from ZIP level values within each county. The categories classification is: Dense ≥300**,** Moderate 100–<300**,** Sparse <100 people/sq mi.

The primary metric (proportional change) that was used is, percent_change = (pop2021 − pop2011) / pop2011

Insight 1: Population change is proportionally different by county density category. Average percent change by category shows a clear gradient: Dense +4.0%**,** Moderate −1.4%**,** Sparse −5.0% (2011 to 2021). This indicates that growth concentrated in denser counties rather than changing proportionally across all categories.



Insight 2: The pattern holds despite variation within each category**.** Ranges show within category variation, but the category level differences remain meaningful:

- Dense: min −3.3% to max +10.0% (avg +4.0%)
- Moderate: min −7.5% to max +5.8% (avg −1.4%)
- Sparse: min −9.7% to max +0.1% (avg −5.0%)

Insight 3: County land area is approximated by summing ZIP land areas (ZIP to county assignment), so density should be interpreted as a consistent proxy rather than an official county area measure.

Appendix:

Question 1 code:

```
WITH county_agg AS (
  SELECT location.county AS county, SUM(acs_estimates.pop2011) AS pop2011, SUM(acs_estimates.pop2021) AS pop2021, SUM(location.area_land_sq_miles) AS land_sq_miles
  FROM location JOIN acs_estimates
  ON acs_estimates.zip = location.zip
  WHERE state = 'PA' AND location.county IS NOT NULL AND location.area_land_sq_miles IS NOT NULL AND location.area_land_sq_miles > 0
  GROUP BY location.county
),
county_classified AS (
  SELECT county, (pop2021 * 1.0 / land_sq_miles) AS density_2021,
    CASE
      WHEN (pop2021 * 1.0 / land_sq_miles) >= 300 THEN 'Dense'
      WHEN (pop2021 * 1.0 / land_sq_miles) >= 100 THEN 'Moderate'
      ELSE 'Sparse'
    END AS density_category,
    (pop2021 - pop2011) * 1.0 / pop2011 AS percent_change
  FROM county_agg
)
SELECT density_category, COUNT(*) AS county_count, ROUND(AVG(percent_change), 4) AS avg_percent_change,ROUND(MIN(percent_change), 4) AS min_percent_change, ROUND(MAX(percent_change), 4) AS max_percent_change
FROM county_classified
GROUP BY density_category
ORDER BY
  CASE density_category WHEN 'Dense' THEN 1 WHEN 'Moderate' THEN 2 ELSE 3 END;
```

Question 2 code:

-- Query A (for bar chart)

```
SELECT

  CASE WHEN (DaysOfTheWeek.DayID IN (1,2,3,4,5)) AND ((HOUR(Rentals.StartTime) BETWEEN 6 AND 9) OR (HOUR(Rentals.StartTime) BETWEEN 16 AND 18)) AND (Rentals.TripDuration_Minutes BETWEEN 3 AND 30) AND (Rentals.FromStationID <> Rentals.ToStationID) AND (Rentals.FromStationID <> -5000) AND (Rentals.ToStationID <> -5000) THEN 'Commutes' ELSE 'Leisure' END AS TypeOfTrip, MetroArea.CoreCity AS MetroArea, HOUR(Rentals.StartTime) AS StartHour, COUNT(*) AS TripCount

FROM Rentals JOIN DaysOfTheWeek

  ON Rentals.StartDayOfWeek = DaysOfTheWeek.DayID

JOIN MetroArea

  ON Rentals.MetroID = MetroArea.MetroID

GROUP BY TypeOfTrip, MetroArea, StartHour

ORDER BY MetroArea, StartHour, TypeOfTrip;
```

-- Query B (for table)

```
SELECT MetroArea.CoreCity AS MetroArea,

  CASE WHEN (DaysOfTheWeek.DayID IN (1,2,3,4,5)) AND ((HOUR(Rentals.StartTime) BETWEEN 6 AND 9) OR (HOUR(Rentals.StartTime) BETWEEN 16 AND 18)) AND (Rentals.TripDuration_Minutes BETWEEN 3 AND 30) AND (Rentals.FromStationID <> Rentals.ToStationID) AND (Rentals.FromStationID <> -5000) AND (Rentals.ToStationID <> -5000) THEN "Commutes" ELSE "Leisure" END AS TypeOfTrip, COUNT(*) AS TripCount

FROM Rentals JOIN DaysOfTheWeek

  ON Rentals.StartDayOfWeek = DaysOfTheWeek.DayID

JOIN MetroArea

  ON Rentals.MetroID = MetroArea.MetroID

GROUP BY MetroArea, TypeOfTrip

ORDER BY MetroArea, TypeOfTrip.
```